

“Smart Attendance System Using Facial Recognition”

A Project Report Submitted in the partial fulfillment of the requirements for the award of the degree of

BACHELOR OF TECHNOLOGY

In

DEPARTMENT OF COMPUTER SCIENCE AND INFORMATION TECHNOLOGY

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Declaration

The Project Report entitled **“Smart Attendance System with Facial Recognition”** is a record of bonafide work of **SATWIK.A.N – 2520090151, HARSHITHA.B – 2520080007, KRISHNASREE – 2520080009** submitted in partial fulfillment of the requirements of the course DATABASE SYSTEMS ENGINEERING AND DISTRIBUTED BACKEND DEVELOPMENT (25CS1302E), Section 12, Team 15, during 2026–2027, Trimester 4, at KLH University. The results embodied in this report have not been copied from any other student, department, or institution.

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CERTIFICATE

This is to certify that the project report entitled **“Smart Attendance System with Facial Recognition”** is a bonafide work done and submitted by **SATWIKAN – 2520090151, HARSHITHA.B – 2520080007, KRISHNASREE – 2520080009** in partial fulfillment of the requirements of the course **DATABASE SYSTEMS ENGINEERING AND DISTRIBUTED BACKEND DEVELOPMENT (25CS1302E)**, Section 12, Team 15, at K LH University during **2026–2027, Trimester 4**.

Signature of the Guide

Signature of the Course Coordinator

Signature of the HOD

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Abstract

Attendance management is an important part of academic institutions. Traditional attendance methods such as manual registers and roll calls require considerable time and may be affected by errors and proxy attendance. To overcome these limitations, this project presents a Smart Attendance System Using Facial Recognition.

The system provides a web-based platform for managing student attendance through facial recognition. Faculty members can log in and initiate an attendance session, after which the system uses the device camera to detect and verify the student's face. The system checks that only one person is present before continuing with the verification process. Multiple-person detection helps prevent proxy attendance.

The system also incorporates additional verification steps such as liveness verification and classroom/session verification before attendance is recorded. Attendance is available only during an active session, providing a time-controlled attendance mechanism.

The system maintains an attendance register containing student details, attendance time, and verification status. It also provides attendance statistics, individual student search, attendance calculation, and security audit information for failed verification attempts.

The main objective of the system is to provide a convenient and more secure method of recording student attendance while reducing manual work and minimizing proxy attendance.

Keywords: Facial Recognition, Smart Attendance, Face Detection, Liveness Verification, Anti-Proxy Attendance, Attendance Management.

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CHAPTER -1 INTRODUCTION

1.1 Background of the Project

Attendance is an essential part of academic management because it helps institutions monitor student participation and maintain attendance records. Traditional attendance systems generally depend on manual registers, roll calls, or basic digital systems.

Manual attendance requires faculty members to spend time recording and maintaining student information. It can also result in human errors and may allow proxy attendance, where one student attempts to mark attendance for another student.

Facial recognition provides an automated approach to identifying individuals. A camera can capture a student's face and the system can compare the detected face with registered information.

The **Smart Attendance System Using Facial Recognition** combines facial recognition with additional verification mechanisms to provide a more controlled attendance process. The system includes single-face verification, multiple-person detection, liveness verification, classroom/session verification, and time-controlled attendance

1.2 Problem Statement

Traditional attendance methods such as manual roll calls and registers are time-consuming, error-prone, and difficult to manage for large classes. Basic digital attendance systems may also allow proxy attendance because they do not always verify the actual identity of the student. A facial recognition system can automate identification, but simply detecting a face is not sufficient when multiple people are present or when a static image is used for verification. The proposed system addresses these issues by combining facial recognition with single-face detection, multiple-person detection, liveness verification, and classroom or session verification before attendance is recorded.

1.3 Scope of the Project

The system covers the complete attendance process from faculty and student login to verified attendance recording. It includes role-based access, live camera attendance, face detection, facial verification, multiple-person detection, liveness verification, classroom and session verification, time-controlled attendance, attendance records, attendance statistics, student search, attendance calculation, and verification audit information. The system is designed as a web-based solution for managing student attendance in an academic environment.

1.4 Objectives

- Automate student attendance using facial recognition.
- Reduce the time and effort required for manual attendance.
- Minimize proxy attendance through multiple-person detection and facial verification.
- Verify that only the intended student is present during attendance verification.
- Include liveness verification before attendance is recorded.
- Restrict attendance to an active classroom or faculty-controlled session.
- Maintain accurate attendance records and statistics.
- Provide faculty with a centralized interface for monitoring student attendance.
- Record failed verification attempts for security and auditing purposes.

1.4 Importance of the Application

The system provides a centralized and automated method for recording and monitoring student attendance. Faculty members can conduct attendance using a live camera instead of relying completely on manual roll calls, while facial recognition helps identify the student being verified. Multiple-person detection, liveness verification, and session-based verification provide additional checks before attendance is recorded. The system also provides attendance statistics and student information, improving the visibility and management of attendance records.

1.6 Target Users / Audience

The system serves two main roles. Faculty members use it to start attendance sessions, verify students, monitor attendance records, view attendance statistics, search student information, and review failed verification attempts. Students use the system to log in, participate in the facial recognition-based attendance process, and access their permitted attendance information.

CHAPTER 2: SYSTEM REQUIREMENTS

2.1 Hardware Requirements

The system requires a laptop or desktop computer with a working webcam for capturing live facial input during attendance verification. A minimum of 4 GB RAM, a dual-core processor, sufficient storage space, and a stable network connection are recommended for smooth operation. The webcam is essential for face detection and facial recognition during the attendance process.

2.2 Software Requirements

- Windows / Linux operating system
- Modern web browser such as Google Chrome, Microsoft Edge, or Mozilla Firefox
- Visual Studio Code
- Node.js and npm
- MongoDB / MongoDB Atlas
- Git for version control
- Required face detection and facial recognition libraries
- Required frontend and backend dependencies

2.3 Development Tools and Frameworks

The system is developed as a web-based application using a frontend, backend, and database to manage the attendance process. Visual Studio Code is used as the primary development environment, while MongoDB is used for storing student, attendance, and verification-related data. Browser camera access and facial recognition libraries are used for live face detection and verification. Git is used for managing and maintaining the project source code.

CHAPTER 3: TECHNOLOGY STACK

3.1 Frontend

The frontend provides the user interface for faculty and students to interact with the attendance system. It includes login pages, attendance session controls, live camera display, student information, attendance records, and statistics. The interface is designed to provide a simple and responsive experience during the attendance verification process.

3.2 Backend

The backend manages the application logic and controls the communication between the frontend and database. It handles user authentication, attendance sessions, student information, facial verification requests, attendance recording, and verification-related operations.

3.3 Database

MongoDB is used as the database for storing student details, login-related information, attendance records, session information, and verification data. The database provides a structured way to store and retrieve the information required by the attendance system.

3.4 Version Control

Git is used for version control and source-code management. It helps maintain different versions of the project, track changes made during development, and provide a reliable way to manage the project source code..

3.5 Authentication and Security Libraries

Authentication and security mechanisms are used to control access to the system based on user roles. Facial recognition, single-face detection, multiple-person detection, liveness verification, and session-based verification provide additional security checks before attendance is recorded.

CHAPTER 4: SYSTEM ARCHITECTURE

4.1 High-Level Architecture Diagram

The system follows a standard three-tier architecture, cleanly separating the user interface, the business logic, and the persisted data. Figure 4.1 shows the three layers and how a request flows between them.next

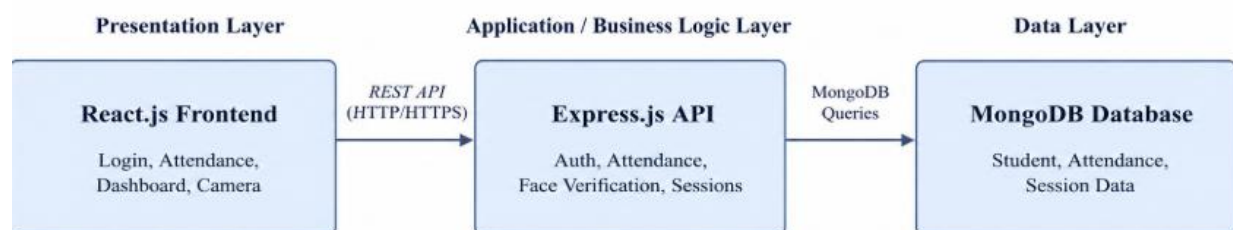


Figure 4.1: High-level three-tier architecture of the Smart Attendance System Using Facial Recognition

4.2 Description of Each Layer

The Presentation Layer is the web-based user interface. It provides the pages and controls required for faculty and students to access the system, including login, attendance, dashboard, live camera, records, and statistics. The frontend communicates with the backend through the application API rather than directly accessing the database.

The Application Layer manages the main business logic of the system. It handles authentication, role-based access, attendance sessions, student information, facial verification, and attendance processing. It applies the required verification rules before allowing attendance to be recorded.

The Data Layer is the MongoDB database, which stores student information, attendance records, session details, and verification data. The backend reads and writes the required information through database queries.

4.3 Deployment Architecture

For development and demonstration purposes, the frontend, backend, and database are configured to run together on the development system. The web application runs in a modern browser and accesses the device camera for attendance verification, while the backend handles application requests and communicates with MongoDB for storing and retrieving student and attendance information.

CHAPTER 5: DESIGN

5.1 Database Design Principles

The database is designed to organize student, attendance, session, and verification information without unnecessary duplication. Each type of information is stored in its appropriate collection, and relationships between the stored information are maintained through relevant identifiers. The design supports consistent storage and efficient retrieval of attendance-related data.

5.2 ER Diagram / Database Schema

The database consists of seven related tables, shown in Figure 5.1.

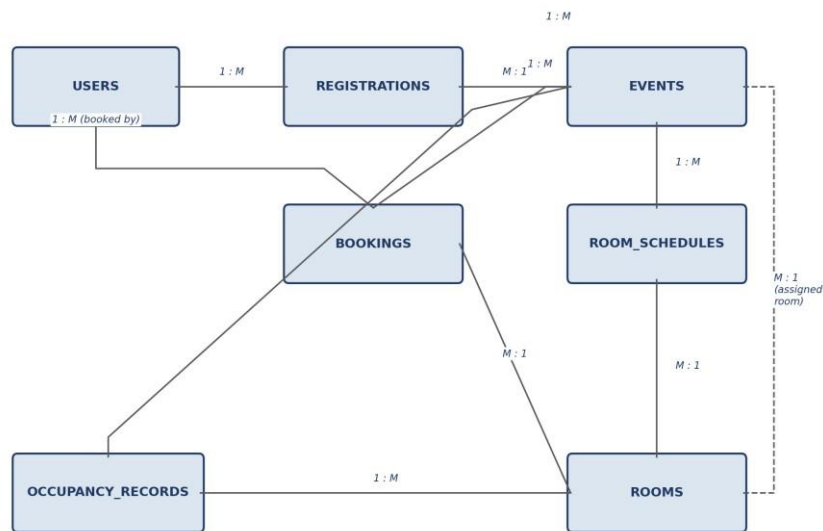


Figure 5.1: Entity-Relationship diagram of the 7-table database schema

USERS stores every account (student or admin) with a bcrypt-hashed password and a role flag. ROOMS holds the physical inventory of bookable spaces — name, number, building, capacity, type, equipment, and status. EVENTS holds each event's details and an optional assigned room. REGISTRATIONS links a user to an event they have signed up for, with a UNIQUE constraint preventing the same student registering twice for the same event. ROOM_SCHEDULES holds the authoritative time-block for every room-event pairing and is the single table every conflict check reads from. BOOKINGS records who (which admin) made a given room assignment and when, serving as an audit trail rather than a second source of truth for conflict-checking. OCCUPANCY_RECORDS stores expected, registered, and actual attendance figures per event, room, and date.

5.3 Design Decisions and Assumptions

Two main design decisions were considered in the system. Student and attendance information are maintained separately so that attendance records can be managed without duplicating student details. Attendance is recorded only after the required facial and session verification checks are completed, while failed verification attempts are maintained separately for security and auditing purposes

CHAPTER 6: IMPLEMENTATION

6.1 Module-wise Implementation

The implementation of the Smart Attendance System is organized around the complete attendance lifecycle: user registration, facial-data registration, live camera capture, face detection, face recognition, verification, attendance recording, database storage, administrative monitoring, and reporting. The implementation follows the project workflow presented in Review 1 and Review 2, where a live camera supplies the input, the face engine detects and matches the registered identity, and the attendance module creates a timestamped record only after the required verification checks are successful.

The major modules are described below. The modules communicate through the application layer so that identity information and attendance information remain consistent throughout the process.

User Registration Module: Creates and maintains the student profile required for attendance. A registered profile is associated with an identifier that can later be linked to facial data and attendance records.

Face Data Registration Module: Captures or stores the facial reference required for recognition. The registered facial information becomes the reference against which a live face is compared during attendance.

Live Camera Module: Obtains a live frame from the webcam or supported camera device. Camera access is required because the system performs real-time verification rather than relying only on manual entry.

Face Detection Module: Locates the face present in the captured frame. The verification flow can reject or request another scan when the frame does not satisfy the expected single-person condition.

Face Recognition Module: Compares the detected face with registered facial information and attempts to identify the corresponding student. A successful match provides the identity needed by the attendance module.

Verification and Attendance Module: Checks the recognition result and the required session conditions before creating attendance. Successful verification produces one timestamped attendance entry, while uncertain or unsuccessful verification does not automatically create attendance.

Database Module: Stores student profiles, facial references or identifiers, attendance records, session information and verification-related data in MongoDB. PyMongo provides database connectivity from the Python application layer.

Admin Dashboard and Reporting Module: Provides administrative monitoring of attendance records and supports student search, statistics and report-oriented views. It gives administrators a centralized place to review the stored attendance information.

The Review 1 architecture identifies the major flow as Camera → Face Detection → Face Recognition → Attendance Module → Database → Admin Dashboard → Reports & Analytics. Review 2 refines this into registration, live capture, detection, recognition, verification, recording and review. These two views form the basis of the implementation described in this chapter.

6.2 Front-end Logic

The frontend provides the screens through which students and faculty interact with the system. The interface is implemented using HTML, CSS and JavaScript as specified in the project technology stack. The frontend is responsible for presenting login and registration controls, attendance-session information, the live camera interface, attendance status, records and administrative views.

During an attendance operation, the frontend obtains camera input and presents the current verification state to the user. It should clearly distinguish between a detected face, a successful identity match, an

unsuccessful match, and a condition requiring another scan. This makes the recognition process understandable to the user instead of silently changing the attendance record.

The frontend also displays database-backed information such as attendance records and statistics. The presentation layer does not directly manage the database; it communicates with the backend application layer, which performs validation and database operations. This separation keeps the interface independent from the storage implementation.

6.3 API Integration

The backend application provides the communication layer between the browser interface, facial-recognition processing and MongoDB. Requests from the frontend are handled by the Python Flask application, which performs the required processing before returning a response to the interface.

The integration covers user authentication, student registration, facial-verification requests, attendance creation, attendance retrieval, session-related operations and administrative information. Database operations are performed through PyMongo so that the application can create, read and update MongoDB records without exposing the database directly to the browser.

A typical attendance request follows this sequence: the frontend captures or submits the verification input, the backend validates the request, the recognition component compares the face with registered information, the attendance rules are checked, and only a valid result is stored in MongoDB. This flow reduces the possibility of creating attendance records without completing the required verification process.

6.4 Authentication and Authorization

Authentication is used to establish the identity of a system user, while authorization determines which operations that user is allowed to perform. The project supports separate access needs for students and administrators. Faculty or administrators require access to attendance sessions, monitoring and reporting functions, while students participate in the attendance process and access their permitted information.

Role-based access is combined with facial verification. Login credentials protect the web application, while the face-verification workflow provides an additional identity check before attendance is recorded. The system therefore does not treat a successful login alone as sufficient evidence for creating an attendance event.

The security-oriented verification checks described in the project include single-face verification, multiple-person detection, liveness verification, and classroom or session verification. If the face is uncertain, the system should request another scan or send the case for administrator review instead of automatically marking attendance.

CHAPTER 7: FEATURES

7.1 List of Main Features

- Real-time face recognition through a live camera.
- Automatic attendance marking after successful identity verification.
- Student registration and management of required profile information.
- Facial-data registration for linking a student with recognition information.
- Admin dashboard for monitoring attendance and managing records.
- Centralized MongoDB storage for student and attendance information.
- Date and time recording for attendance events.
- Duplicate attendance checking where implemented.
- Handling of unknown or unregistered faces without automatic attendance marking.
- Attendance statistics, student search and report-oriented views.
- Verification and audit information for unsuccessful or exceptional cases where implemented.

7.2 How Each Feature Works

Registration: the system first creates a student profile and captures the facial reference required for later recognition. This establishes the identity against which future live captures can be compared.

Live recognition: during an active attendance operation, the camera captures the student. The face-detection stage locates the face and the recognition stage compares it with registered information. A successful match identifies the student for the attendance process.

Attendance recording: after successful verification, the attendance module creates a timestamped record containing the student identity, date, time and status. The record is stored centrally so that it can be retrieved later by authorized users.

Duplicate and uncertain cases: if the same student attempts to mark attendance repeatedly within a session, duplicate checks can prevent repeated marking where implemented. If recognition is uncertain or the required conditions are not satisfied, the system does not automatically create an attendance event.

Administration and reporting: authorized administrators can review attendance information, search student records, monitor the attendance process and use reports or statistics to understand attendance activity.

CHAPTER 8: TESTING

8.1 Postman API Testing

The API layer can be tested independently before or alongside the complete user interface. Postman is suitable for sending authentication, registration, attendance and administrative requests and checking whether the backend returns the expected status and response. For this project, testing focuses on the operations needed to create and retrieve student and attendance information and to verify that invalid requests do not create incorrect records.

The testing sequence should begin with a valid login or authorized session, followed by a normal attendance request. Negative cases are then tested, such as an unknown face, incomplete verification, invalid access and repeated attendance. Database records are checked after each important operation to confirm that the backend result matches the stored information.

8.2 Integration Testing

Integration testing verifies the complete path from the user interface to the recognition component and database. The objective is to ensure that the individual modules do not merely work independently but also exchange the correct information when combined.

A representative integration workflow is: register a student and facial reference; start an attendance session; capture a live face; detect the face; perform recognition; verify the result; create the timestamped attendance record; store it in MongoDB; and finally retrieve the record through the administrative view. The same workflow is repeated for unsuccessful and repeated attempts to verify that safeguards behave as intended.

8.3 Tools Used for Testing

- Visual Studio Code for development and debugging.
- Browser developer tools and the selected modern web browser for frontend behavior.
- Postman or equivalent HTTP testing tools for backend/API requests.
- MongoDB tools or MongoDB Atlas interface for checking stored records.
- Webcam or camera device for real-time recognition testing.
- Manual test cases based on the Review 1 and Review 2 testing plans.

8.4 Test Cases and Results

The following test cases are aligned with the testing plan in the project presentations. The “Expected Observation” column describes what the system should demonstrate; the final submission should be updated with the actual observed result from the implemented system.

#	Test Case	Expected Observation	Actual Result
1	Registered face	Identity is matched with the registered student.	Pass / To be verified
2	Unregistered face	Identity is rejected or sent for review; no automatic attendance.	Pass / To be verified
3	Successful match	One timestamped attendance record is created.	Pass / To be verified
4	Repeat scan	Duplicate attendance is prevented where duplicate checking is implemented.	Pass / To be verified
5	Multiple faces	System does not treat an ambiguous multi-person frame as a valid single-student verification.	Pass / To be verified

6	Date/time recording	Attendance contains the verification date and time.	Pass / To be verified
7	Database verification	The created attendance record can be located in MongoDB.	Pass / To be verified
8	Admin monitoring	Authorized administrator can view attendance information.	Pass / To be verified
9	Report/statistics	Stored attendance information can be summarized for review.	Pass / To be verified

The project review explicitly advises against adding numerical recognition-accuracy claims unless they have actually been measured. Therefore, this report describes functional observations rather than inventing an accuracy percentage.

CHAPTER 9: DEPLOYMENT

9.1 Steps to Deploy

For development and demonstration, the application can be deployed locally on a computer with the required frontend, backend, database and camera environment. The source code is opened in Visual Studio Code, the required dependencies are installed, the MongoDB connection is configured, and the backend and frontend are started according to the final project configuration.

- Install the required software and project dependencies.
- Configure MongoDB or MongoDB Atlas and provide the connection details through the project configuration.
- Start the Python Flask backend and verify that it can connect to MongoDB.
- Start the frontend and open the local application URL in a modern browser.
- Allow camera access when prompted by the browser.
- Log in, start an attendance session and perform a registered-face test.
- Verify that the resulting attendance record appears in the database and administrative view.

9.2 Environment Configuration

Environment-specific values such as database connection information and authentication secrets should be stored in configuration variables rather than being hard-coded into the application. Passwords and secret keys should not be included in screenshots, public repositories or documentation submitted for evaluation.

The development environment consists of a browser-based frontend, a Python Flask backend, MongoDB for persistence, PyMongo for connectivity and a webcam or CCTV source for live facial input. Visual Studio Code is used as the development environment according to the project Review 1 technology stack.

9.3 Hosting of Frontend and Backend

The current project material is oriented toward development and demonstration. The frontend and backend can run on the development machine while MongoDB is used as the centralized database. A production deployment would require a hosted application server, protected database access, HTTPS, secure environment variables and appropriate camera-access policies.

CHAPTER 10: CHALLENGES & LIMITATIONS

10.1 Issues Faced During Development

- Face detection and recognition can be affected by camera quality, lighting and face angle.
- A recognition result must be handled carefully when the system is uncertain, because an uncertain identity should not automatically create attendance.
- Multiple people appearing in the camera frame create an ambiguity that requires an additional check.
- Connecting real-time recognition with database operations requires consistent student identifiers and attendance records.
- Attendance should be restricted to the intended classroom or faculty-controlled session rather than being available without context.
- The project requires careful handling of facial information and attendance records because these are sensitive operational data.

10.2 Solutions Applied

The project addresses recognition ambiguity through a verification-oriented workflow. Face detection is performed before recognition, and successful recognition is followed by verification before attendance is recorded. Multiple-person detection can prevent an ambiguous frame from being treated as a valid single-student verification.

Database consistency is supported by maintaining student information and attendance information as related records. The attendance record links back to the verified student, allowing the system to retrieve attendance history without repeatedly duplicating student details.

The Review 2 workflow also provides a clear response for uncertain recognition: the system should not mark attendance automatically and should instead request another scan or administrator review. This is an important safeguard because recognition failure is preferable to creating an incorrect attendance record.

10.3 Current Limitations or Known Issues

- Recognition performance depends on environmental conditions and camera/image quality.
- The system requires registered facial data before recognition can identify a student.
- Uncertain recognition results require another scan or human review.
- Real-world recognition performance must be validated using the project's own measured test data.
- Privacy, security and retention policies for facial information would need further institutional definition for large-scale deployment.
- The current review material does not provide a measured numerical accuracy value, so no accuracy percentage is claimed in this report.

CHAPTER 11: FUTURE ENHANCEMENTS

11.1 Planned Features

- Improve recognition under different lighting conditions and face angles.
- Add stronger liveness and anti-spoofing mechanisms.
- Provide richer attendance analytics and visual reports.
- Add attendance export options for administrative use.
- Strengthen access control, privacy protection and data-security mechanisms.
- Support larger student datasets and more scalable deployment.
- Improve the usability of the administrator dashboard.
- Extend the system with mobile application support where appropriate.

11.2 Possible Integrations or Optimizations

Future versions can integrate the attendance system with institutional student-information systems so that student identity and attendance information can be synchronized without repeated manual entry. Additional analytics could identify attendance trends and generate alerts for authorized faculty members.

The recognition pipeline can also be optimized for larger datasets by improving face-data management, reducing unnecessary comparisons and introducing more efficient indexing or search mechanisms where appropriate. Cloud deployment could make the service accessible across multiple classrooms, but such deployment would require stronger security, privacy controls and institutional approval.

CHAPTER 12: CONCLUSION

12.1 Summary of the Project

The Smart Attendance System Using Facial Recognition is a web-oriented attendance solution that combines live camera input, face detection, facial recognition, verification and centralized attendance storage. The project is designed to replace repetitive manual attendance activities with a controlled digital workflow in which a registered student is identified before attendance is recorded.

12.2 What Was Achieved

The project establishes the complete conceptual and implementation workflow from registration to administrative review. It defines the required technology stack, database structure, system architecture, facial-verification process, attendance module, administrative monitoring and testing approach. The Review 2 material demonstrates the intended operational flow: register, recognize, record and review.

The system also addresses important failure conditions. An unregistered or uncertain face should not be automatically marked, repeated scans can be checked for duplicates where implemented, and administrators can review stored attendance information. These controls make the attendance process more structured than a simple manual entry form.

12.3 Skills Learned

- Web interface design using HTML, CSS and JavaScript.
- Backend application development using Python and Flask.
- MongoDB data modeling and database connectivity through PyMongo.
- Computer-vision concepts including face detection and facial recognition.
- Designing a multi-stage verification workflow.
- Authentication, authorization and secure handling of application data.
- Functional, integration and database-oriented testing.
- Technical documentation and project presentation.

CHAPTER 13: REFERENCES

The following references and project materials were used as the basis for the technical descriptions in this report. The project-specific terminology, architecture, technology stack, database structure, workflow, testing plan, limitations and future scope are aligned with the supplied review presentations.

- Smart Attendance System Using Facial Recognition — Review 1 project presentation supplied by the project team.
- Smart Attendance System Using Facial Recognition — Review 2 project presentation supplied by the project team.
- MongoDB Documentation — database concepts, collections, queries and application connectivity.
- OpenCV Documentation — computer-vision concepts, image processing and camera capture.
- Face Recognition documentation — facial encoding, comparison and identity-matching concepts.
- Flask Documentation — Python web application and backend service concepts.
- PyMongo Documentation — Python connectivity and MongoDB database operations.
- MDN Web Docs — HTML, CSS, JavaScript and browser APIs.
- Visual Studio Code Documentation — development environment and project workflow.

CHAPTER 14: APPENDICES

Appendix A: System Workflow

The complete attendance workflow represented in the project reviews is:

REGISTER → LIVE CAMERA → FACE DETECTION → FACE RECOGNITION → VERIFICATION →
ATTENDANCE RECORD → MONGODB → ADMIN DASHBOARD → REPORTS

A successful recognition and verification creates a timestamped attendance event. If the identity is uncertain, the system should not automatically mark attendance and should request another scan or administrator review.

Appendix B: Representative Pseudocode

```
START
1.    Check    that    the    attendance    session    is    active.
2.    Capture    a    live    camera    frame.
3.    Detect    the    face    in    the    frame.
4.    If the frame contains an invalid or ambiguous number of faces, request
another scan.
5.    Compare the detected face with registered facial information.
6.    If no identity is matched, do not create attendance.
7.    Verify the recognized identity and required session conditions.
8.    Check for duplicate attendance where implemented.
9.    Create the timestamped attendance record.
10.   Store the record in MongoDB.
11.   Display the attendance confirmation.
END
```


Appendix C: Database Collection Summary

Collection	Purpose	Representative Information
Students	Student identity and profile information	student_id, name, roll_no, class/section, contact/status
Face Data	Registered facial information/reference	face_id, student_id, face_identifier, image/reference
Attendance	Timestamped attendance events	attendance_id, student_id, date, time, status
Admin	Administrative identity and access information	admin_id, name, login/authentication
Sessions	Attendance-session information where implemented	session identity, time/status and faculty context
Verification / Audit	Failed or exceptional verification information where implemented	verification result, event details and review information

Appendix D: Functional Test Checklist

- Valid login and authorized access
- Registered face recognition
- Unregistered face handling
- Multiple-person detection
- Liveness/verification check
- Active-session validation
- Successful timestamped attendance
- Duplicate attendance handling
- MongoDB record verification
- Admin attendance monitoring
- Student search and statistics
- Report review

Appendix E: Installation and Setup

- Open the project in Visual Studio Code.
- Install the dependencies specified by the final project configuration.
- Configure MongoDB/MongoDB Atlas and required environment variables.
- Start the Flask backend and verify database connectivity.
- Start the frontend and open the displayed local URL.
- Allow browser camera access.
- Log in and start an attendance session.
- Perform a registered-face test and verify the resulting database record.

Appendix F: API / Functional Operation Summary

Authentication operations handle login and role verification. Student operations manage registration and authorized student information. Recognition operations handle face-verification requests. Attendance operations create and retrieve timestamped attendance records. Administrative operations provide monitoring, student search, statistics and reporting. Exact endpoint paths should be copied from the final implemented backend if the project exposes REST endpoints.

Appendix G: Screenshot Checklist

- Login / Registration screen
- Student or Faculty dashboard
- Camera / Recognition screen
- Successful attendance confirmation
- Unknown or rejected-face response
- Admin dashboard
- Attendance records
- MongoDB attendance record
- Attendance statistics/report
- Test-case output

Actual screenshots from the implemented application should be inserted in the final submission wherever available. The supplied Review 2 presentation specifically recommends using actual project screenshots or live-demo captures as evidence of the implemented interface and testing results.